

Full Stability Sweep Report — 2026-06-19

Field	Value
Run date	2026-06-19 02:15 UTC
Project	github.com/HelixDevelopment/helix_ota
Server commit	HEAD of main
Gate script	tests/pre_build_verification.sh
Report path	docs/qa/20260619-full-stability-sweep/REPORT.md

1. Server Go Test Suite

Command: `cd server && go test -count=1 -v ./...`

Result: ALL TESTED PACKAGES PASS

Package	Result	Time
server/internal/api	PASS	0.384s
server/internal/config	PASS	0.344s
server/internal/deviceemu	PASS	1.163s
server/internal/fabric	PASS	0.595s
server/internal/health	PASS	0.458s
server/internal/rollout	PASS	0.723s
server/internal/store	PASS	0.854s
server/internal/transport	PASS	1.227s
server/internal/api/manager-dist	(no tests)	—
server/cmd/ota-device-emu	(no tests)	—
server/cmd/ota-server	(no tests)	—
server/tools/loadtest	(no tests)	—

Evidence: `/tmp/server_tests_r2.log`

2. Stress + Chaos Tests

Command: `bash tests/stress_chaos/run_server_stress.sh`

Result: PASS (exit code 0)

Phase	Subtest	Result	Key metrics
Stress	TestStressConcurrentGroupCreate	PASS	p50=271us p95=797us p99=1.02ms
Stress	TestStressSustainedReads	PASS	p50=10us p95=350us p99=1.20ms
Stress	TestStressConcurrentMembershipNoLostUpdates	PASS	
Stress	TestStressConcurrentAuth	PASS	10/10 2xx
Stress	TestStressConcurrentDevice	PASS	10/10 2xx
Stress	TestStressConcurrentRelease	PASS	1 created / 99 conflicts (expected)
Chaos	TestChaosAuthBadPayload (4 cases)	PASS	All rejected 401
Chaos	TestChaosAuthHugePayload	PASS	401 graceful degradation
Chaos	TestChaosConcurrentMutation	PASS	49/50 conflicts (expected)
Chaos	TestChaosStoreRestart	PASS	Baseline 201 -> fault 500 -> recovered 201
Chaos	TestChaosRepoFaultDegradesAndRecovers	PASS	200 -> 500 -> 200

Evidence dir: qa-results/stress_chaos/20260618T231559Z/

3. Pre-Build Gates

Command: bash tests/pre_build_verification.sh

Result: PASS (exit code 0)

Gate	Result	Notes
constitution-inheritance A (clean tree)	PASS	All 7 invariants met
constitution-inheritance B (sec1.1 mutation)	PASS	Mutation correctly caused FAIL
	PASS	

Gate	Result	Notes
constitution-inheritance C (submodule pointers)	PASS	All 8 owned submodules wired
helixqa-bank-runner-self- test A (missing evidence)		Correctly FAILED
helixqa-bank-runner-self- test B (real evidence)		Correctly PASSEd
CM- COVENANT-114-153 through 158 PROPAGATION	OK	All 6 propagation gates pass

Note: First run FAILED because the constitution submodule was not checked out. After `git submodule init` && `git submodule update`, all gates pass clean.

Evidence: `/tmp/gates_r2.log`

4. Go Submodule Tests

Command: `for sm in ota-protocol ota-artifact-validator ota-rollout-engine ota-telemetry-schema http3; do cd submodules/$sm && go test -count=1 ./...; done`

Result: ALL 5 SUBMODULES PASS

Submodule	Result	Time
submodules/ota-protocol	PASS	0.221s
submodules/ota-artifact-validator	PASS	0.249s
submodules/ota-rollout-engine	PASS	2.354s
submodules/ota-telemetry-schema	PASS	0.195s
submodules/http3	PASS	1.111s

5. Go Build + Vet

Step	Exit Code	Result
go build ./...	0	PASS
go vet ./...	0	PASS

Note: First run FAILED because the containers submodule was not checked out. After `git submodule update`, both build and vet pass.

Overall Verdict

Check	Result
Server test suite (7 packages with tests)	PASS
Build	PASS
Vet	PASS
Stress tests	PASS
Chaos tests	PASS
Pre-build gates	PASS
Go submodule tests (5 submodules)	PASS
OVERALL	PASS

Anomalies noted

1. **Submodule initialization required:** The constitution and containers submodules were not checked out. A fresh clone requires `git submodule update --init` before tests/gates pass.
2. **Containers dependency:** The `digital.vasic.containers` replace directive in `go.mod` blocks all packages importing `internal/deviceemu` or `internal/transport` when the submodule is missing.
3. **No tests for cmd packages:** `ota-device-emu`, `ota-server`, `manager-dist`, and `loadtest` have no test files (expected for CLI entry points).
4. **Concurrent release conflicts:** `TestStressConcurrentRelease` reports 99 conflicts from 100 concurrent version bumps – this is the expected optimistic-concurrency behaviour.